

MULTI-YEAR ANALYSIS OF ELECTRIFIED STORM CELLS OVER NORTH ALABAMA

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INTRODUCTION AND METHODOLOGY

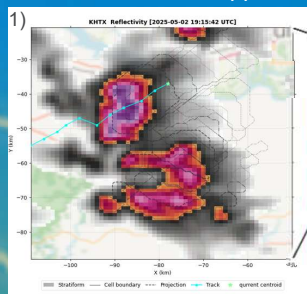
- While lightning is an objectively gorgeous phenomenon to observe, the number of flashes over the US had increased by 20% from 2024 to 2025 (Vaisala 2025). Kingfield 2017 showed an increase in lightning strikes to wind turbines compared to other tall structures while Steiger 2007 showed how lightning activity could be used to indicate severe storm growth on a faster update interval than conventional weather radars.
- This study makes use of the WSR-88D radar in Huntsville, AL (KHTX) alongside the North Alabama Lightning Mapping Array (NALMA).
 - KHTX is a dual-polarization radar which scans hydrometeors horizontally and vertically to get products like differential reflectivity and more alongside reflectivity, velocity, etc. (Ex. Fig. 1 and Fig. i)
 - The NALMA network is a set of time of arrival sensors which detect radiation emitted by lightning. The sensors are then able to compute the lat/lon/alt/time of the radiation source, providing a view of the evolution of a flash. (Ex. Fig. 2 and Fig. ii)
- North AL was chosen due to the proximity to an Atmospheric Research Monitoring (ARM) site in the nearby Bankhead National Forest (BNF).
- Analysed a particular event on May 2-3rd, 2025 due to the high volume of storm cells compared to other events in the dataset.
- Adapt is primarily an orchestrated framework for total analysis of a storm cell where the unique characteristics of a cell can be analysed (Raut et al. 2026). Xlma-python was used to process the LMA data and characterize flashes, providing the ability to correlate individual flashes with storm cells and relate characteristics of both (Bruning). (Ex. Fig. 3)

Fig. 1) Reflectivity plot of a few storm cells over North Alabama captured by KHTX doppler radar for the time period starting at 19:15:42Z on 05-02-2025.

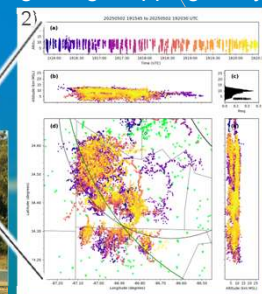
Fig. 2) LMA plot of lightning sources captured by the NALMA network. Data captured is shown in a) time-altitude, b) longitude-altitude, c) frequency-altitude, d) longitude-latitude, and e) altitude-latitude. Rings represent radii from the center of the LMA network and KHTX radar. Green shapes represent towers defined by the FAA. The plotted data is for the same time as shown in the radar scan.

Fig. 3) Plot of LMA sources overlap of radar reflectivity, showing how different regions of a cell correlate to different lightning regions.

Combining Storm Observations from NEXRAD Doppler Radar



With High Resolution Lightning Data from Lightning Mapping Array



To get a better understanding of electrified storm cells

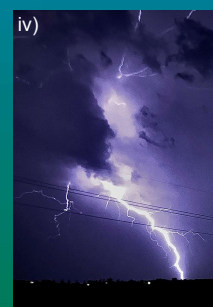
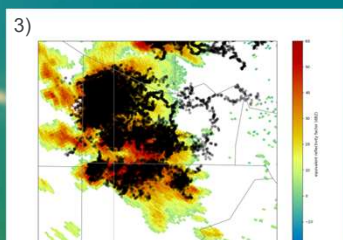


Photo by Landon Moeller

Photo by Bee Lamsma

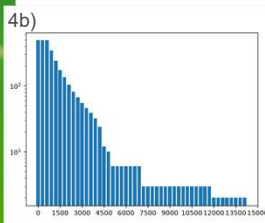
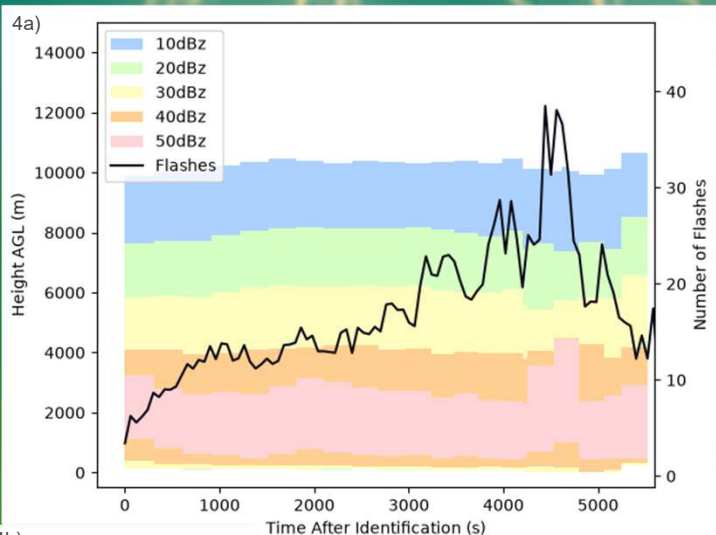


Fig. 4a) Bar plot showing the average max and min echo top heights at every 10 dBZ interval for every cell above listed threshold recorded during the event with average number of lightning flashes at the same period in the cell's lifespan overlaid. The cells were captured by KHTX in Huntsville, AL in a 150 km radius. Lightning data was captured by the NALMA, centered on Huntsville, in a 200 km radius. All the data in the following figures is from the same region.

4b) Bar chart showing the distribution of cells on a logarithmic scale for the time following identification. The first few scans include ~500 cells but by 3000 s the count drops to ~50 and by 6000 s the count is down to 5 scans..

RESULTS AND FINDINGS

- Event stats:** 288 radar scans, ~2500 storm cells, and 120431 lightning flashes, 943 electrified storm cells, longest lived cell lasted 11074 s (3 hr 5 min), longest flash lasted 2.62 s.
- Filtering:** cells over 10 km², at least 3 scans, and did not have a merge in the lifespan, lightning sources with a χ^2 of 1.0, detected by >6 stations, flashes separated by >3 km and >0.15 s, and >5 sources.
- Figure 4a shows an average of every cell in the filtered dataset plotted with average number of flashes at the same time index. 4b shows the number of cells for each time index which drastically drops after 3000 s. The peaks in reflectivity and number of flashes is likely due to a low sample size from cells that were strong enough to live that long.
- Figure 5 shows a comparison between the highest measured point of a cell and the highest LMA source for the same cell. The points above the line show when there was lightning above the cloud which, even considering a maximum 2 km elevation error range (Chmielewski and Bruning 2017), there exist flashes extending above the recorded cloud top.
- ZDR and dBZ values suggest heavy rain or hail in the 2-4 db region where the flash rate peaks. This region represents the possibility of moderate updrafts with large rain drops or small-medium hail lofted within, which would support increased noninductive charging and a higher flash rate.

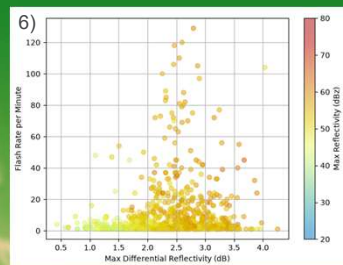
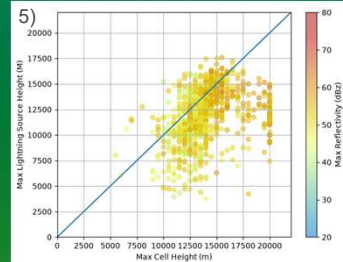


Fig. 5) Scatter plot comparing the maximum height of a cell in a radar scan with the associated highest recorded LMA source. Points are colored by the maximum reflectivity of the cell taken from the 0.5° elevation scan.

Fig. 6) Plot showing the distribution of lightning rates for every cell's 0.5° elevation scan max differential reflectivity. Points colored the same as Fig. 5.

CONCLUSIONS

- This study proves that there is value in looking at this data over a larger time period to gain a better understanding of how lightning relates to cells across the seasons.
- A larger sample size will be needed to generalize the shown characteristics, but for this event:
 - Lightning rates appeared to continue increasing throughout the cell's lifecycle even when the reflectivity core did not grow.
 - Flash area, source count, flash rate, and flash length were all plotted against max ZDR with slight variations in where the plot peaked representing how different lightning characteristics are impacted by changes in a single radar variable..
- While this study did very little analysis that has not been performed in other studies, the application of Adapt in combination with an LMA network has allowed for the possibility to more easily manage multi-year datasets for larger studies.

FURTHER WORK

- First priority is to expand this project to analyze radar and lightning data across the 7 years that the NALMA network has been active.
- Making use of ARM-BNF sensors to gain a better understanding of how electrified storm characteristics relate to other processes during a storm's lifecycle, using various sensors to get an idea of the storm's microphysics.
- Perform the same analysis over West Texas, Houston, Oklahoma, Colorado, and Langmuir LMA networks to see trends across different storm environments.
- Future applications of Adapt include continuing to maintain as a real-time forecast tool and additional modules made for different areas of atmospheric science.
 - Ability to correlate ASOS data with an overhead cell, storm reports for hail/tornado identification, GLM for regions where LMA is not available, etc.

ACKNOWLEDGMENTS

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- To Eric Bruning and others in the Texas Tech Lightning Research Group for the creation and maintaining of pylma which was vital to process the LMA data used.
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- To the DOE Office of Science SULI program for the funding and Argonne National Lab for hosting me.
- Further work regarding the subject and references are available upon request to beelamsma@outlook.com